

ITEP 310

CHAPTER 2: PROFESSIONALISM IN IT



WHY PROFESSIONALISM MATTERS IN IT

1. IT as a Trust-Based Profession

- IT professionals often handle sensitive information: personal data, financial transactions, health records, and intellectual property.
- Clients, employers, and society trust IT workers to protect this information.
- Without professionalism, trust is broken, leading to data breaches, financial losses, and damaged reputations.



Example: A careless IT administrator who leaves a server unpatched could cause a massive security breach, costing millions and putting customer privacy at risk.

WHY PROFESSIONALISM MATTERS IN IT

2. Ethical Decision-Making in Technology

- Technology evolves faster than laws. Many situations IT professionals face have no clear legal guidance, but require ethical judgment.
- Professionalism ensures that decisions are guided by fairness, integrity, and responsibility, not just profit.

Example: An app developer discovers their product collects more data than necessary. Legally it may be allowed, but professionalism requires **respecting user privacy** and minimizing data collection.



WHY PROFESSIONALISM MATTERS IN IT

3. Accountability to Multiple Stakeholders

Professionalism means balancing responsibilities to:

- **Yourself** – Stay competent and honest.
- **Employer/Clients** – Provide reliable solutions, protect assets.
- **Society** – Ensure technology serves the public good and avoids harm.



Example: An IT consultant who discovers a system flaw must decide between hiding it (to please the employer) or reporting it (to protect society). Professionalism ensures the **right choice** is made.

WHY PROFESSIONALISM MATTERS IN IT

4. Professionalism Builds Career Reputation

- A strong professional image makes IT workers more employable and respected.
- Employers value professionals who:
 - Communicate clearly
 - Deliver on time
 - Follow ethical standards
- Professionalism also helps in global collaboration, since IT teams often work across borders.



Example: An IT specialist with a reputation for reliability and ethical standards is more likely to receive promotions, leadership roles, or international opportunities.

WHY PROFESSIONALISM MATTERS IN IT

5. Professionalism Prevents Misuse of Technology

- IT tools are powerful – they can be used for progress or harm.
- Professionalism ensures that systems are built for security, inclusivity, and fairness.
- Without professionalism, IT can contribute to: Cybercrime, AI Bias, Privacy Violations, Social Inequality



Example: Facial recognition systems can be biased if poorly designed. A professional IT team ensures **diverse testing and ethical use before deployment.**

IT CODES OF CONDUCT

What Are IT Codes of Conduct?

A **code of conduct** is a **set of ethical principles and professional guidelines** created by organizations to govern the behavior of IT professionals.

Purpose:

- **Ensure responsible use of technology**
- **Provide guidance when facing ethical dilemmas**
- **Protect clients, employers, and society**



MAJOR IT CODES OF CONDUCT

- 1. ACM** (Association for Computing Machinery)
- 2. IEEE** (Institute of Electrical and Electronics Engineers)
- 3. ISACA** (Information Systems Audit and Control Association)



MAJOR IT CODES OF CONDUCT

1. ACM (Association for Computing Machinery)

What it is:

- **The world's largest educational and scientific computing society, founded in 1947.**

Who it serves:

- **Software developers, computer scientists, IT researchers, students.**

Main role:

- a. Promotes computing as a science and profession.**
- b. Publishes journals, conferences, and research in computer science.**
- c. Provides a Code of Ethics to guide computing professionals.**

Example:

If a programmer is designing an app, ACM's code guides them to avoid harm, respect privacy, and ensure fairness.

MAJOR IT CODES OF CONDUCT

2. **IEEE** (Institute of Electrical and Electronics Engineers)

What it is:

- The world's largest technical professional organization, founded in **1963**, covering engineering, electronics, and IT.

Who it serves:

- Engineers, IT professionals, innovators in AI, robotics, telecommunications, and more.

Main role:

- a. Creates standards (e.g., Wi-Fi, Bluetooth, 802.11).
- b. Publishes journals and runs conferences.
- c. Provides a Code of Ethics that prioritizes public safety, honesty, and lifelong learning.

Example:

An engineer working on self-driving cars must follow IEEE's principle of public safety above company profit

MAJOR IT CODES OF CONDUCT

3. **ISACA** (Information Systems Audit and Control Association)

What it is:

- A global professional association for IT governance, auditing, risk management, and cybersecurity, founded in 1969.

Who it serves:

- IT auditors, security specialists, risk managers, compliance officers.

Main role:

- a. Provides certifications like CISA, CISM, CRISC, CGEIT.
- b. Sets frameworks for IT governance and risk management.
- c. Offers a Code of Professional Ethics focusing on confidentiality, integrity, and due care.

Example:

An IT auditor working for a bank follows ISACA's code to ensure customer financial data remains confidential and secure

MAJOR IT CODES OF CONDUCT

Organization	Focus Area	Who It Serves	Key Ethics Emphasis
ACM	Computing, software, research	Programmers, computer scientists, IT students	Fairness, privacy, avoiding harm
IEEE	Engineering, technology standards	Engineers, technologists, innovators	Public safety, honesty, responsibility
ISACA	IT governance, auditing, cybersecurity	IT auditors, risk managers, security experts	Confidentiality, integrity, competence

THE IMPORTANCE OF COMMUNICATION IN IT

1. Why Communication Matters in IT

- IT professionals don't just deal with machines — they work with clients, managers, teammates, and end-users.
- Clear communication ensures:
 - a. **Understanding** – Explaining technical issues in simple terms.
 - b. **Efficiency** – Avoiding project delays caused by miscommunication.
 - c. **Trust** – Building credibility with clients and employers.
 - d. **Collaboration** – Enabling teamwork across different roles (developers, analysts, users).

Example: If a programmer uses too much jargon when explaining a bug, a manager may not understand its urgency. Clear, simple communication ensures everyone is on the same

THE IMPORTANCE OF COMMUNICATION IN IT

2. Types of Communication in IT

- a. **Technical Communication** - Writing documentation, system manuals, project reports.
- b. **Interpersonal Communication** - Speaking with clients, teammates, and supervisors.
- c. **Nonverbal Communication** - Professional tone, body language, presentation style.
- d. **Cross-Cultural Communication** - Working with international teams where culture and language differ.

WHY PROFESSIONALISM MATTERS IN IT

3. Significance in Key IT Situations

a. Client Communication

- IT professionals must explain solutions in non-technical terms so clients understand value and risks.

Example: Explaining “data encryption” as “locking information with a digital key only authorized users can open.”

b. Team Communication

- Large IT projects require collaboration between developers, designers, testers, and managers.
- Miscommunication can cause delays, errors, or security risks.

WHY PROFESSIONALISM MATTERS IN IT

3. Significance in Key IT Situations

c. Crisis Communication

- During system failures or cyberattacks, IT professionals must communicate quickly and clearly with stakeholders to minimize panic and damage.

d. Cross-Cultural Teams

- IT projects often involve international outsourcing and multicultural teams.
- Professionals need sensitivity to cultural differences in tone, hierarchy, and expectations.



Thanks for playing!

THE END!

